

A sharp finite-regime lower bound for random palindrome blocks

Research proof; two independent full reviews completed

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Write $R_\sigma = n!P(\sigma)$ for m independent uniformly relabelled palindrome blocks on n labels. This note completes the lower bound for this random family, including the logarithm, by comparing small assignment problems with Gaussian ones. It concerns this family only.

Theorem 1. *There are absolute constants $c, C > 0$ and n_0 such that, for $n \geq n_0$ and*

$$m \geq Cn^{13/10}(\log n)^{1/5},$$

$$\mathbb{P}\left(\max_{\sigma \in S_n} |R_\sigma - 1| \geq c \frac{n^{13/4} \sqrt{\log n}}{m^{5/2}}\right) \geq 1 - Cn^{-9/10}.$$

Thus the previously proved sufficient scale $n^{13/10}(\log n)^{1/5}\varepsilon^{-2/5}$ matches the typical necessary scale for this construction in the large-block, small-error regime.

We use the notation and elementary estimates from the companion notes. Set

$$d = n-3, \quad w_i(a) = \binom{d}{a} p_i^a (1-p_i)^{d-a}, \quad p_i = (i-1)/(m-1), \quad h = \binom{n}{3} m^{-3}(1-1/m)^d.$$

The triple projection is $S_\sigma = h \sum_i \sum_a w_i(a) \delta_{i,a}$, with $\delta(l, x, r) = \frac{1}{2} - \frac{3}{2} \mathbf{1}\{x < \min(l, r)\}$. For central starts $a \in [d/4, 3d/4]$ and $m \geq n$,

$$\max_i w_i(a) \leq Cn^{-1/2}, \quad cmn^{-3/2} \leq \sum_i w_i(a)^2 \leq Cmn^{-3/2}. \quad (1)$$

These bounds follow from Stirling, the beta integral, and the Riemann-sum estimate for a unimodal function, as detailed in *A matching power lower bound for random palindrome blocks*.

1 A smooth comparison inequality

Lemma 2. *Let X_i be independent centered random vectors in $\mathbb{R}^D$, and let independent centered Gaussian vectors G_i have the respective same covariance matrices. For $\beta > 0$,*

$$\left| \mathbb{E} F_\beta \left(\sum_i X_i \right) - \mathbb{E} F_\beta \left(\sum_i G_i \right) \right| \leq C \beta^2 \sum_i (\mathbb{E} \|X_i\|_\infty^3 + \mathbb{E} \|G_i\|_\infty^3),$$

where $F_\beta(x) = \beta^{-1} \log \sum_{\alpha=1}^D e^{\beta x_\alpha}$. Also $\max_\alpha x_\alpha \leq F_\beta(x) \leq \max_\alpha x_\alpha + \beta^{-1} \log D$.

Proof. For any direction v , the third directional derivative of F_β is β^2 times the third centered moment of the numbers v_α under the softmax probabilities. Its absolute value is at most $8\beta^2 \|v\|_\infty^3$. Replace the vectors one at a time and expand along the replaced vector through degree two. The first two Taylor terms cancel because means and covariance matrices match. The stated third-order remainder bound proves the lemma. The maximum inequalities are immediate. $\square$

2 One small assignment problem

Put $L = \lfloor n^{1/10} \rfloor$. Inside a window of $3L + 4$ consecutive central positions, choose L movable positions spaced by three, leaving two positions of padding at each end. Keep the other labels fixed and permute the L movable labels arbitrarily among the movable positions. There are $D = L!$ choices π . Exactly three triple starts around each movable position can be affected. Their set $\mathcal{A}$ has size $3L$, and all corresponding triples lie inside this window.

Let T_π be the contribution of these starts to S under assignment π , and normalize by

$$s_0 = h \sqrt{m/n^{3/2}}, \quad X_i(\pi) = \frac{h}{s_0} \sum_{a \in \mathcal{A}} w_i(a) \delta_{i,a}^\pi.$$

Then $T_\pi/s_0 = \sum_i X_i(\pi)$, a sum of independent centered vectors. Write $M_i = \max_{a \in \mathcal{A}} w_i(a)$. From (1),

$$\sum_i M_i^3 \leq \sum_{a \in \mathcal{A}} \sum_i w_i(a)^3 \leq CL \frac{m}{n^2}. \quad (2)$$

The deterministic bound on a single vector is

$$\|X_i\|_\infty \leq CL \frac{n^{3/4}}{\sqrt{m}} M_i.$$

For each fixed target, valley covariances vanish beyond distance two and have uniformly bounded absolute row sums. Thus

$$\max_{\pi} \text{Var } X_i(\pi) \leq CL \frac{n^{3/2}}{m} M_i^2.$$

A Gaussian vector with coordinate variances at most v^2 satisfies $\mathbb{E}\|G\|_\infty^3 \leq Cv^3(\log(2D))^{3/2}$, by the Gaussian tail bound and a union bound. Since $\log(2D) \leq CL \log L$, (2) gives

$$\sum_i (\mathbb{E}\|X_i\|_\infty^3 + \mathbb{E}\|G_i\|_\infty^3) \leq CL^4(\log L)^{3/2} \frac{n^{1/4}}{\sqrt{m}}. \quad (3)$$

The Gaussian vector $G = \sum_i G_i$ contains an independent assignment matrix. Specifically, let $u(l, x, r)$ be the canonical three-variable Hoeffding component of δ , for which $\mathbb{E}u^2 = 1/20$. For each movable position v and movable label b , project onto

$$\frac{h}{s_0} (w_i(v-1)u(Y_{i,\lambda_v}, Y_{i,b}, Y_{i,\rho_v}))_{i=1}^m,$$

where λ_v, ρ_v are the two fixed adjacent anchors. Distinct such kernels are orthogonal, since their triples share at most two variables and u is centered given any two. The corresponding Gaussian entries $Z_{v,b}$ are independent, and their variances satisfy

$$\text{Var } Z_{v,b} = \frac{h^2}{20s_0^2} \sum_i w_i(v-1)^2 \asymp 1$$

by (1). Exactly as in the companion Gaussian-limit note,

$$G(\pi) = \sum_v Z_{v,\pi(v)} + H(\pi),$$

with the centered Gaussian residual independent of the matrix. No limiting argument is used here: this is an orthogonal decomposition of the finite Gaussian surrogate with the given covariance matrix. Conditional Jensen and a greedy row-by-row assignment therefore give

$$\mathbb{E} \max_{\pi} G(\pi) \geq cL\sqrt{\log L}. \quad (4)$$

Indeed the first $L/2$ rows each have at least $L/2$ fresh independent Gaussian columns with standard deviations bounded below, and later greedy increments have nonnegative expectations.

Apply Lemma 2 with $\beta = B\sqrt{\log L}$, where a sufficiently large absolute B makes the smoothing error $(\log D)/\beta$ less than one fourth of (4). The comparison error, divided by $L\sqrt{\log L}$, is at most

$$CL^3(\log L)^2 \frac{n^{1/4}}{\sqrt{m}} \leq Cn^{-1/10}(\log n)^2 = o(1)$$

whenever $m \geq n^{13/10}$. Therefore, uniformly in this range,

$$\mathbb{E}(\max_{\pi} T_{\pi} - \min_{\pi} T_{\pi}) \geq cs_0 L \sqrt{\log L}. \quad (5)$$

We used only $\mathbb{E} \min_{\pi} T_{\pi} \leq 0$, since each coordinate is centered.

We also need a second-moment upper bound. For any fixed π ,

$$\sum_i \text{Var } X_i(\pi) \leq CL, \quad |X_i(\pi)| \leq CLn^{1/4}/\sqrt{m}.$$

The elementary Bernstein moment estimate for a sum of bounded independent centered variables gives, for $q \geq 2$,

$$\left\| \sum_i X_i(\pi) \right\|_q \leq C \left(\sqrt{qL} + qLn^{1/4}/\sqrt{m} \right).$$

This estimate follows by expanding each exponential moment, summing the geometric remainder using boundedness, and integrating the resulting Bernstein tail bound. Take $q = \max(2, \log(2D))$ and use $\|\max_{\pi} |X(\pi)|\|_q \leq D^{1/q} \max_{\pi} \|X(\pi)\|_q$. Since $L \leq n^{1/10}$ and $m \geq n^{13/10}$, this yields

$$\mathbb{E}(\max_{\pi} T_{\pi} - \min_{\pi} T_{\pi})^2 \leq Cs_0^2 L^2 \log L. \quad (6)$$

3 Independent windows and the full distribution

Fit $Q \asymp n/L$ disjoint windows of the above type in the middle third of a base target. Permute their movable labels independently. The family $\mathcal{F}$ has $(L!)^Q$ members, at most $e^{n \log n}$. Every affected triple lies wholly in one window, so the range of the triple projection over this family is exactly the sum of the Q individual ranges. These ranges are independent because their

label sets are disjoint in every block. From (5), (6), and Chebyshev, with probability $1 - O(Q^{-1})$,

$$\max_{\sigma \in \mathcal{F}} |S_\sigma| \geq cQs_0L\sqrt{\log L} \geq c \frac{n^{13/4}\sqrt{\log n}}{m^{5/2}}.$$

Here $h \asymp n^3/m^3$, $\log L \asymp \log n$, and $Q^{-1} = O(n^{-9/10})$.

Put $A = n^{13/4}\sqrt{\log n}/m^{5/2}$ and $t = m/n$. The coefficient estimates in the refined upper-bound note, with $p = 4n \log n$, give for $U_\sigma = R_\sigma - 1 - S_\sigma$

$$\|U_\sigma\|_p \leq C_1 A(A + t^{-1})$$

for sufficiently large n and $m \geq C_0 n^{13/10}(\log n)^{1/5}$. For completeness, short-support levels $\ell \geq 2$ contribute $O(A^2)$; the short-support first level of lengths $k \geq 4$ contributes $O(A/t)$; the long-support tilted estimate is $\sqrt{2} \exp(-n \log t/12 + 4pn/t^4) \leq A/t$. These are the same coefficient-level estimates used in the earlier power-only lower bound, with the larger value of p .

A union bound over $\mathcal{F}$ and Markov at threshold $eC_1 A(A + t^{-1})$ fail with probability at most $e^{-3n \log n}$. Taking the absolute threshold constant C_0 large enough makes that remainder at most half of the triple-projection lower bound, because $A \leq C_0^{-5/2}$ and $t^{-1} \leq C_0^{-1}$. This proves Theorem 1.

A threshold consequence. For some fixed absolute $\varepsilon_0 > 0$, if $m = o(n^{13/10}(\log n)^{1/5})$, the probability of relative ε_0 -fairness tends to zero. To see this, when m is at least linear in n , restrict to k labels with $m \asymp Ck^{13/10}(\log k)^{1/5}$ and apply the theorem; then $k \rightarrow \infty$, $k \leq n$, and the lower error is an absolute constant. For $m < n/(4e)$, outcome counting already forces an absent permutation. The remaining cases have $m \rightarrow \infty$ and admit the same restriction. Relative fairness is preserved by restriction because marginal relative errors are averages of full-order relative errors.

Dependencies and scope. The elementary binomial estimates and full nonlinear remainder estimates are proved in the companion notes *A refined uniform bound for random palindrome blocks* and *A matching power lower bound for random palindrome blocks*. The canonical variance $1/20$ and Gaussian assignment decomposition are proved in *The logarithmic factor in the large-block Gaussian limit*. Here that decomposition is applied to a finite covariance-matched Gaussian surrogate, and Lemma 2 supplies the quantitative step. No lower bound for general approximate dice constructions follows.